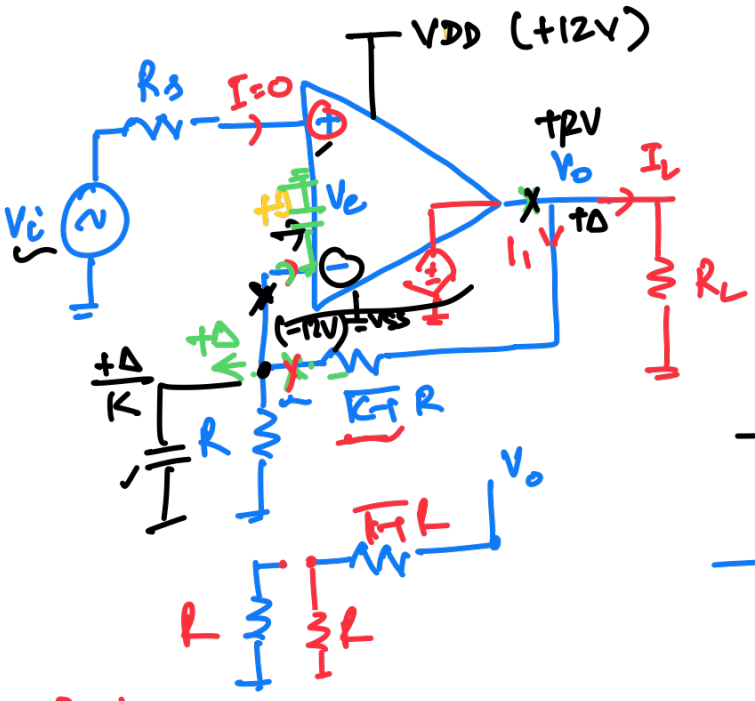
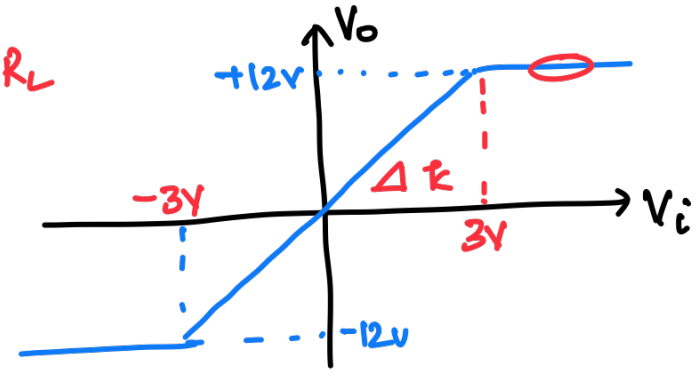
Lecture # 05

$V_o = A V_e$

$$\frac{V_o}{V_i} = \frac{k}{1 + \frac{k}{A}}\Bigg|_{A \to \infty} = k$$

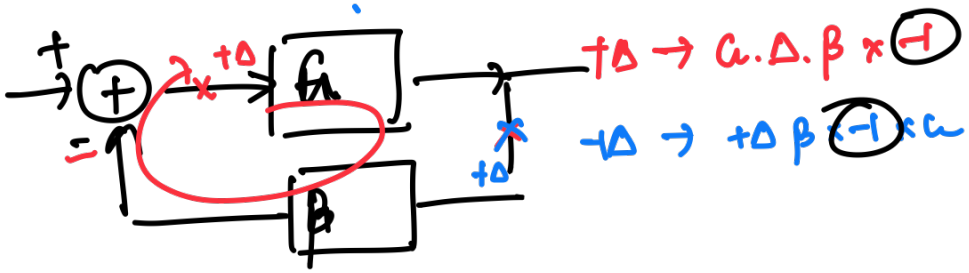




DC Characteristics

- $k = 4$
- $V_i \in [-3, +3V]$, $V_+ = V_- = V_i$
- $V_i = +5V$, V_-

$V_o = +12V$, $V_- = \frac{12}{4} = 3V$, $V_+ = 5V$ $\Rightarrow V_e = \pm 2V \xrightarrow{\times A} A \times 2V$

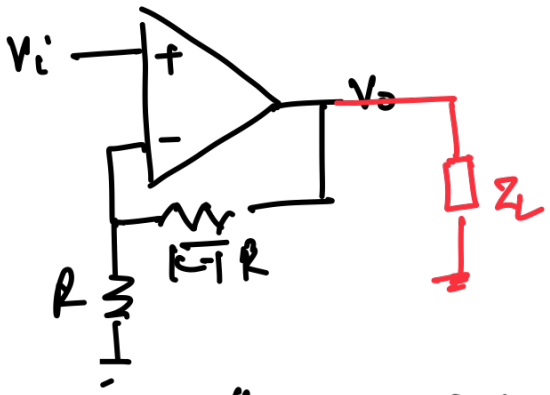


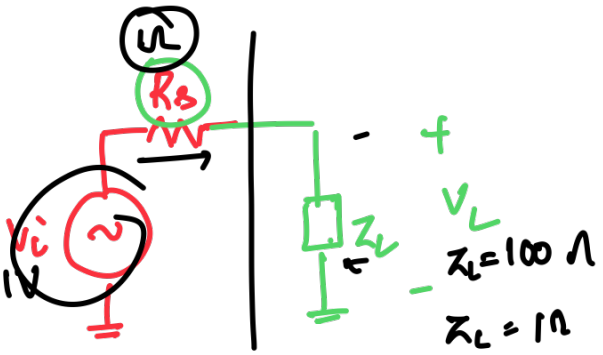
$+\Delta \longrightarrow +\Delta \times \frac{1}{k} \times -1 \times A$

$+\Delta \longrightarrow +\Delta \times -1 \times A \times 1$

$+\Delta \longrightarrow +\Delta \times -1 \times A \times \frac{1}{k}$

"Non-inverting Amp."

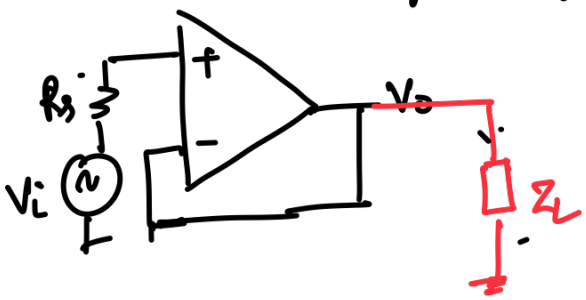




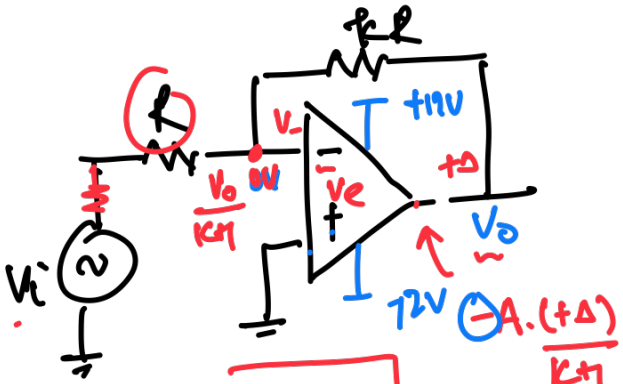
$V_L = V_i$ =

$V_L = \frac{Z_L}{R_s + Z_L} \cdot V_i$

"Voltage-Buffer"



$V_o = V_i$

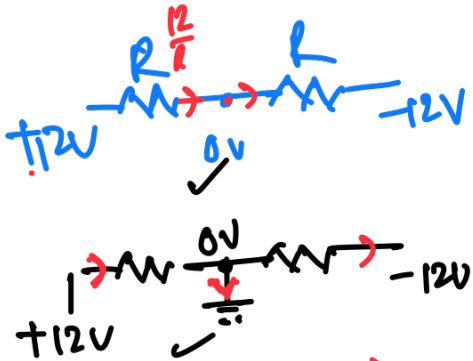


$V_o = A V_e$

- $A \to \infty \Rightarrow V_+ = V_- \Rightarrow V_- = 0V$

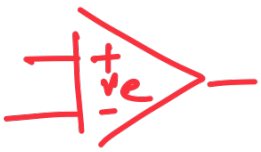
$\frac{V_i - 0}{R} = \frac{0 - V_o}{KR}$

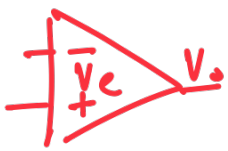
$\frac{V_o}{V_i} = -K$



A is finite

$V_- = V_i \times \frac{K}{K+1} + V_o \cdot \frac{1}{1+K}$







$V_o = A V_e$

$V_e = V_+ - V_-$

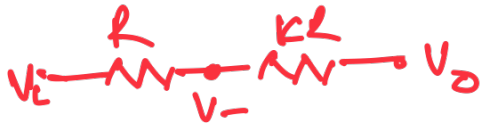
$$V_- = V_i \times \frac{k}{k+1} + V_o \times \frac{1}{k+1}$$

$$V_e = V_+ - V_- = 0 - \left(V_i \frac{k}{k+1} + \frac{V_o}{k+1}\right)$$

$$V_o = AV_e = -A\left(V_i \frac{k}{k+1} + \frac{V_o}{k+1}\right)$$

$$V_o\left(\frac{1}{A} + \frac{1}{k+1}\right) = -V_i \frac{k}{k+1}$$

$$V_o = \frac{-kV_i}{\left(1 + \frac{k+1}{A}\right)}\Bigg|_{A\to\infty} = -kV_i$$

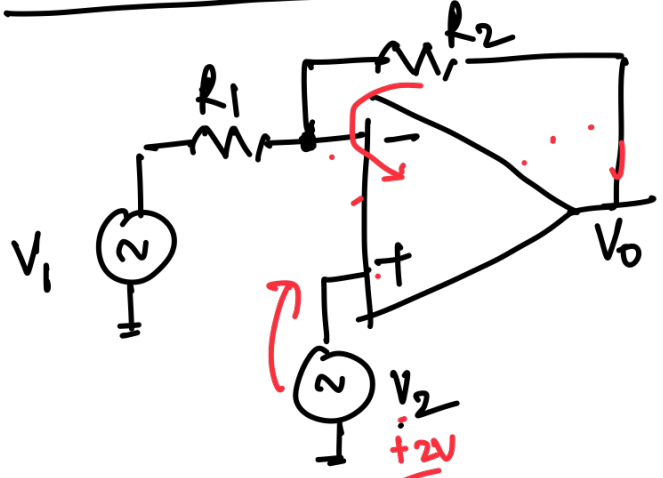


$$V_- = \frac{kR}{(k+1)R} V_i + \frac{k}{(k+1)R} V_o$$

$$V_- = V_i \times \frac{k}{k+1} + \frac{-kV_i}{(k+1)\left(1 + \frac{k+1}{A}\right)}$$

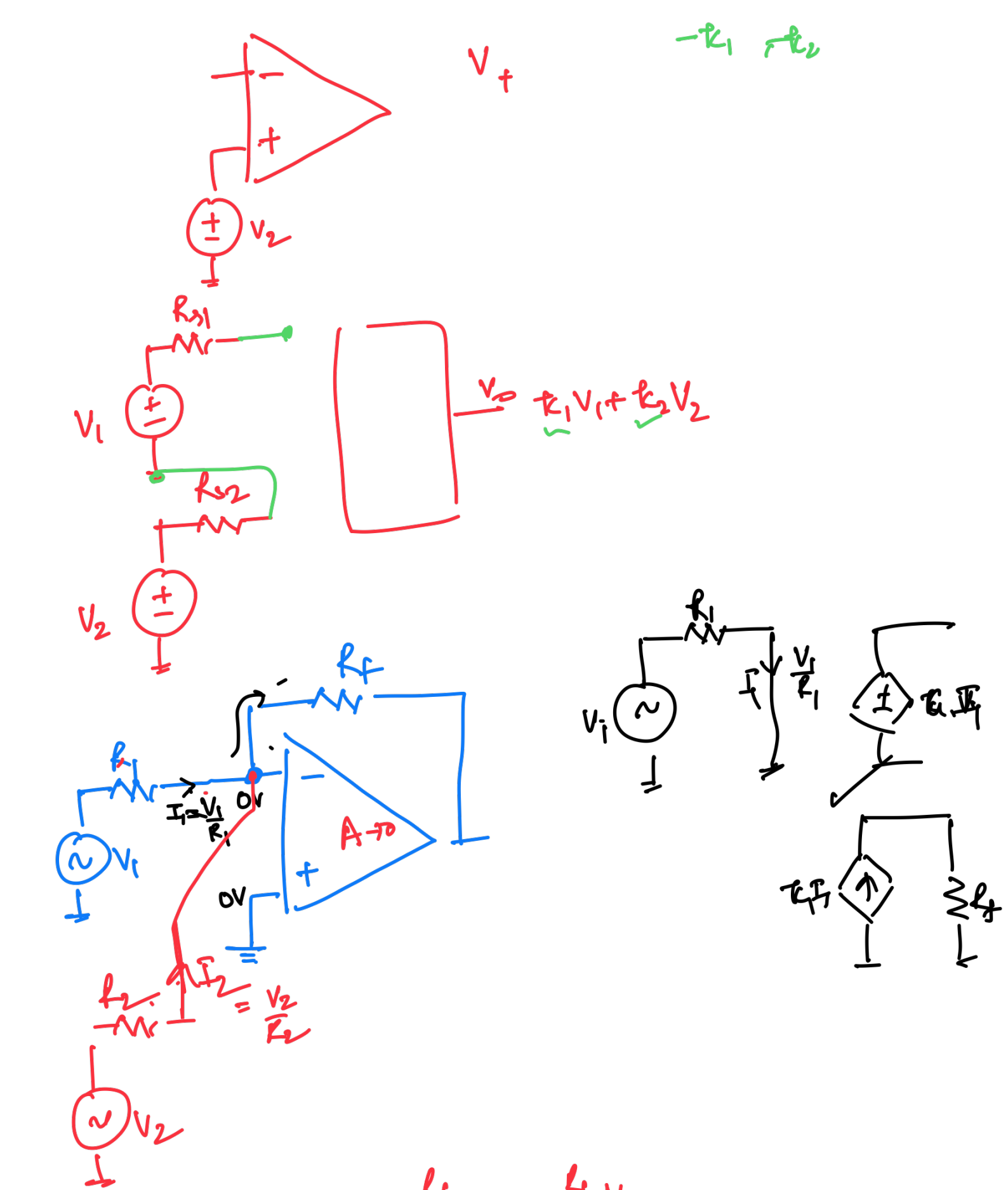
$$= \frac{k}{k+1}\left(1 + \frac{k+1}{A} - 1\right) V_i \times \frac{1}{\left(1 + \frac{k+1}{A}\right)} = \frac{+V_i \cdot k/A}{1 + \frac{k+1}{A}}$$

$$= \frac{+V_i}{\frac{A}{k}\left(1 + \frac{k}{A} + \frac{1}{A}\right)} = \frac{+V_i}{\frac{A}{k} + 1 + \frac{1}{k}}$$



- $V_+ = V_2$, $V_o = AV_e$
- $A \to \infty \Rightarrow V_+ = V_- = V_2$
- $\frac{V_1 - V_2}{R_1} = \frac{V_2 - V_o}{R_2}$

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_2 - \frac{R_2}{R_1} V_1$$



$$V_o = -\frac{R_f}{R_2}V_2 - \frac{R_f}{R_1}V_1$$

$$\frac{R_f}{R_2} \sim k_2 \qquad \frac{R_f}{R_1} \sim k_1$$